数理分析方法

课程导论与证明入门

Lecture 000

原课件逐页保留 · 零基础中文讲解 · 证明训练

材料与编写说明

课程原页：张顺明教授课件（用户提供）。参考：Gérard Debreu, Theory of Value (1959)，第一章 Mathematics。原页与本讲义的补充、纠错、例题分开呈现。本文为辅助自学材料，不代表教师官方讲义。

怎样阅读

先看本页的中文说明，再对照原页符号；遇到证明，先抄出假设和目标，再逐行说明“这一步用的是什么”。遮住证明重写一遍，最后用反例检查条件是否必要。重复或动画展开页仍逐页保留；说明会指出新增步骤或对应的完整论证位置。

“补充”用于补基础或展开原课件略去的论证；“纠错”指出原页需要修正的地方。课程概览及人物介绍页按其原有用途解释，不虚构数学定理。

编写日期：2026 年 10 月 24 日

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零基础读证明：随手查阅

对象的层次。 $x \in A$ 表示一个元素属于集合，$A \subseteq B$ 表示 A 的每个元素都在 B 中。$\{x\}$ 是只含 x 的集合，不是 x 本身。$\varnothing$ 是空集，$\{\varnothing\}$ 则含有一个元素。$\mathbb{R}^N$ 中的点是有 N 个坐标的有序列表。

量词。 $\forall$ 是“对所有”，$\exists$ 是“至少存在一个”。例如“每个学生有一本书”不要求是同一本；“有一本书供每个学生使用”则要求同一本。证明极限时 $\forall \varepsilon > 0\ \exists n_0\ \forall n \geq n_0$ 的顺序不可换：先指定精度，再找统一起点，最后控制这个起点之后的全部项。

怎样否定。 否定“所有 x 满足 P”得到“存在 x 不满足 P”；否定“存在”得到“所有都不”。否定 $P \Rightarrow Q$ 是 P 真且 Q 假。因此要推翻一个定理，必须满足其全部假设又违反结论；拿不满足假设的例子不能当反例。

四种基本证明。 (1) 包含 $A \subseteq B$：任取 $x \in A$，用定义得到 $x \in B$。(2) 相等 $A = B$：两个方向分别证明。(3) 等价 $P \iff Q$：证明 $P \Rightarrow Q$ 及反向。(4) 反证：把结论否定加入已知，推出矛盾。逆否命题 $\neg Q \Rightarrow \neg P$ 与 $P \Rightarrow Q$ 等价，“反过来” $Q \Rightarrow P$ 则未必成立。

数列、子列和收敛。 数列 x_n 是第 n 个对象；子列 x_{n_k} 的指标必须严格递增。$x_n \to x$ 指任意小的误差最终都能保持。前面有限多项不影响极限。抽子列通常是为了用有界性和有限维 Bolzano–Weierstrass 定理获得极限；抽取之后须把原来所有相关量一起限制到同一子列。

距离的关键估计。 $\|x\| = (\sum_i x_i^2)^{1/2}$，$\|x - y\|$ 是距离。三角不等式 $\|x - z\| \leq \|x - y\| + \|y - z\|$ 允许经中间点拆误差。两段误差各小于 $\varepsilon/2$，总误差便小于 ε；分成 m 段时可各分配 ε/m。这里的 ε 不是固定的小数，而是任意正精度。

做证明题的五行草稿。 先写完整假设；写准确目标；列出可用定义与定理；逐步注明所用条件；最后检查任意性、存在性和量词顺序。出现“显然”时问：若阅卷者要求展开，我能否补出一个集合等价链、一个不等式或一个明确构造？不知道证明的命题要标为引用，不能把待证结论循环当作理由。

1 原课件第 1 页：课程与学习目标

Mathematical Analysis Methods

Professor Shunming Zhang

School of Finance
Renmin University of China

September 10, 2022

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

这门课用数学语言研究经济均衡。“存在”问的是满足全部条件的对象能否找到，不一定要求算出它。先掌握集合、逻辑和极限，再学习连续性、凸性和不动点；不必提前学习微积分。证明题的基本格式是：写假设，写目标，选任意对象，逐步调用定义，最后回到目标。

2 原课件第 2 页：参考文献如何使用

Mathematical Analysis Methods

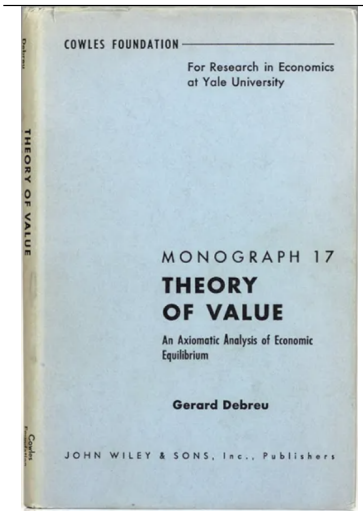
1. Gerard Debreu (1959), **Theory of Value — An Axiomatic Analysis of Economic Equilibrium**
2. Gerard Debreu (1982), *Existence of Competitive Equilibrium*. In Kenneth J. Arrow and Michael D. Intriligator (1982), **Handbook of Mathematical Economics (Volume 2)**, Chapter 15, Pages 697 - 743.
3. Gerard Debreu (1983), **Mathematical Economics** — Twenty Papers of Gerard Debreu.
4. Claude Berge (1959), **Espaces Topologiques**. Translated as **Topological Spaces** (1963)

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

四项文献是课程的知识来源，不是四份额外作业。本讲义以课件为主，用 Debreu (1959) 第一章核对术语。Berge 对应连续性和最优化工具，后续章节使用这些工具研究经济均衡。不要把阅读历史书目等同于证明训练。

3 原课件第 3 页：教材七章的逻辑

Gerard Debreu (1959)



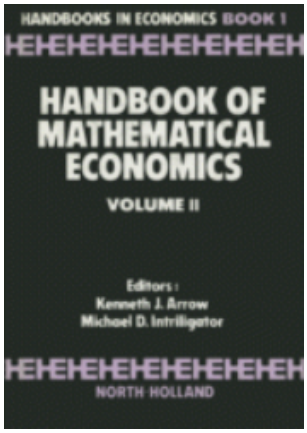
1. *Mathematics*
2. *Commodities and Prices*
3. *Producers*
4. *Consumers*
5. *Equilibrium*
6. *Optimum*
7. *Uncertainty*

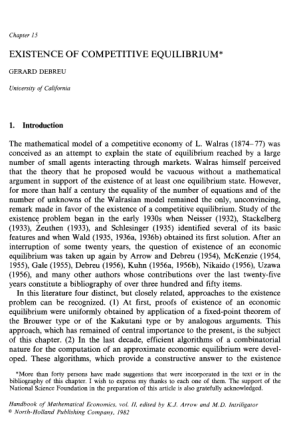
上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

商品和价格把经济对象表示为向量；生产者和消费者描述可行集合与偏好；均衡要求各方选择相容；最优讨论效率；不确定性扩大商品的描述。这十二讲主要负责第一章数学工具，后六章在此只作动机。

4 原课件第 4 页：两种均衡研究路径

Gerard Debreu (1982)





Chapter 15

EXISTENCE OF COMPETITIVE EQUILIBRIUM*

GERARD DEBREU

University of California

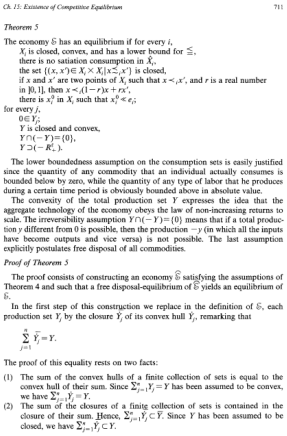
1. Introduction

The mathematical model of a competitive economy of L. Walras (1874–77) was conceived as an attempt to explain the state of equilibrium reached by a large number of small agents interacting through markets. Walras himself perceived that the theory that he proposed would be vacuous without a mathematical argument in support of the existence of at least one equilibrium state. However, for more than half a century the equality of the number of equations and of the number of unknowns of the Walrasian model remained the only, unconvincing, remark made in favor of the existence of a competitive equilibrium. Study of the existence problem began in the early 1930s when Neisser (1932), Stackelberg (1933), Zeuthen (1933), and Schlesinger (1935) identified several of its basic features and when Wald (1935, 1936a, 1936b) obtained its first solution. After an interruption of some twenty years, the question of existence of an economic equilibrium was taken up again by Arrow and Debreu (1954), McKenzie (1954, 1955), Gale (1955), Debreu (1956), Kuhn (1956a, 1956b), Nikaido (1956), Uzawa (1956), and many other authors whose contributions over the last twenty-five years constitute a bibliography of over three hundred and fifty items.

In this literature four distinct, but closely related, approaches to the existence problem can be recognized. (1) At first, proofs of existence of an economic equilibrium were uniformly obtained by application of a fixed-point theorem of the Brouwer type or of the Kakutani type or by analogous arguments. This approach, which has remained of central importance to the present, is the subject of this chapter. (2) In the last decade, efficient algorithms of a combinatorial nature for the computation of an approximate economic equilibrium were developed. These algorithms, which provide a constructive answer to the existence

*More than forty persons have made suggestions that were incorporated in the text or in the bibliography of this chapter. I wish to express my thanks to each one of them. The support of the National Science Foundation in the preparation of this article is also gratefully acknowledged.

Handbook of Mathematical Economics, vol. II, edited by K.J. Arrow and M.D. Intriligator
© North-Holland Publishing Company, 1982



Ch. 15: Existence of Competitive Equilibrium 711

Theorem 5

The economy $\mathcal{E}$ has an equilibrium if for every i,
X_i is closed, convex, and has a lower bound for $\trianglelefteq_i$,
there is no satiation consumption in $\hat{X}_i$,
the set $\{(x, x') \in X_i \times X_i | x \prec_i x'\}$ is closed,
if x and x' are two points of X_i such that $x \prec_i x'$, and r is a real number in $]0,1[$, then $x \prec_i (1-r)x + rx'$,
there is x_i^0 in X_i such that $x_i^0 \ll e_i$,
for every j,
$0 \in Y_j$,
Y_j is closed and convex,
$Y_j \cap (-Y_j) = \{0\}$,
$Y \supset (-R_+^l)$.

The lower boundedness assumption on the consumption sets is easily justified since the quantity of any commodity that an individual actually consumes is bounded below by zero, while the quantity of any type of labor that he produces during a certain time period is obviously bounded above in absolute value.

The convexity of the total production set Y expresses the idea that the aggregate technology of the economy obeys the law of non-increasing returns to scale. The irreversibility assumption $Y \cap (-Y) = \{0\}$ means that if a total production y different from 0 is possible, then the production $-y$ (in which all the inputs have become outputs and vice versa) is not possible. The last assumption explicitly postulates free disposal of all commodities.

Proof of Theorem 5

The proof consists of constructing an economy $\bar{\mathcal{E}}$ satisfying the assumptions of Theorem 4 and such that a free disposal-equilibrium of $\bar{\mathcal{E}}$ yields an equilibrium of $\mathcal{E}$.

In the first step of this construction we replace in the definition of $\mathcal{E}$, each production set Y_j by the closure $\bar{Y}_j$ of its convex hull $\bar{Y}_j$, remarking that

$$\sum_{j=1}^{n} \bar{Y}_j = Y.$$

The proof of this equality rests on two facts:

(1) The sum of the convex hulls of a finite collection of sets is equal to the convex hull of their sum. Since $\sum_{j=1}^{n} Y_j = Y$ has been assumed to be convex, we have $\sum_{j=1}^{n} \bar{Y}_j = Y$.

(2) The sum of the closures of a finite collection of sets is contained in the closure of their sum. Hence, $\sum_{j=1}^{n} \bar{Y}_j \subset \bar{Y}$. Since Y has been assumed to be closed, we have $\sum_{j=1}^{n} \bar{Y}_j \subset Y$.

1. Introduction
2. The Simultaneous Optimization Approach
3. The Excess Demand Approach
4. Economies with A Measure Space of Agents

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

同时最优化强调各人的最优选择能同时实现；超额需求方法用“需求减供给”表示市场失衡。连续性和不动点是存在论证的桥梁。测度空间代理人属于更高级材料，本讲无需提前掌握测度论。

5 原课件第 5 页：均衡思想的来源

Introduction of General Equilibrium

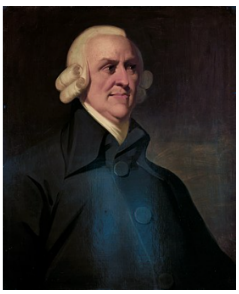
1. Adam Smith (1776), **The Wealth of Nations**
   Invisible Hand = Equilibrium Price
2. Leon Walras (1874-1877), **Elements of Pure Economics**
   Walrasian Equilibrium
3. Kenneth J. Arrow and Gerard Debreu (1954), *Existence of an Equilibrium for a Competitive Economy*, Econometrica 22, 265 - 290
4. Gerard Debreu (1959), **Theory of Value – An Axiomatic Analysis of Economic Equilibrium**

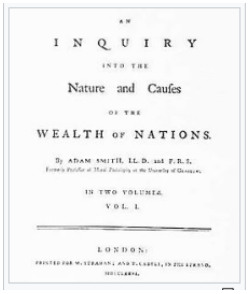
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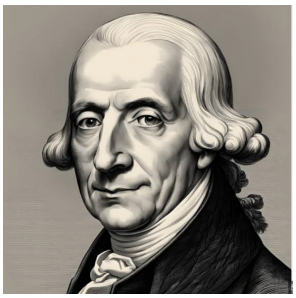
“看不见的手”是直觉性表述，数学均衡需要列出价格、需求、供给及市场出清等条件。Walras 和 Arrow–Debreu 的工作把问题写成可检验的数学命题。不能从历史口号直接推出任何经济都必有均衡，必须检查模型假设。

6 原课件第 6 页：Adam Smith

Introduction







Adam Smith (1776), **The Wealth of Nations**
June 5 (OS) / June 16 (NS) 1723 - July 17, 1790

1. Scottish moral philosopher
2. Pioneering political economist

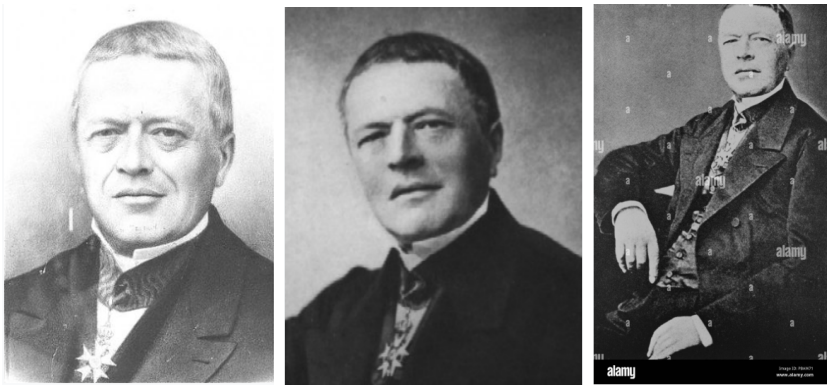
1. The Theory of Moral Sentiments (1759)
2. An Inquiry into the Nature and Causes of the Wealth of Nations (1776)

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页介绍人物、日期和著作。对证明考试，重点是理解经济问题的动机，不把人物年表当成数学推理。形式化之后，“某种价格能够协调选择”才成为待证明的存在命题。

7 原课件第 7 页：Cournot

Introduction



Antoine Augustin Cournot (1838), **The Recherches**

August 28, 1801 - March 31, 1877

French economist, philosopher and mathematician

Researches into the Mathematical Principles of the Theory of Wealth (1838)

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

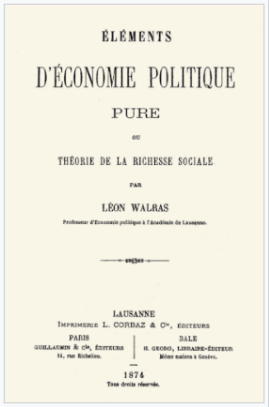
Cournot 把相互影响的选择放进数学模型。一个简单动机是两家企业分别选择产量，各自利润取决于两家的选择。一个人的最优选择可能是集合，这解释了为什么后面除了函数还需要“对应”。

8 原课件第 8 页：Walras

Introduction



Walras in 1862 (age 27)





Marie-Esprit-Leon Walras (1874-1877),

Elements of Pure Economics

December 16, 1834 - January 5, 1910

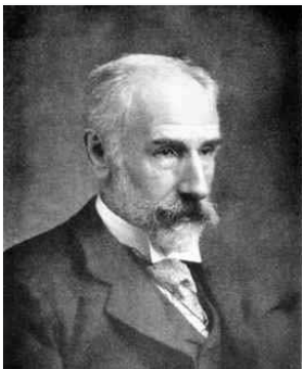
Mathematical economist associated with the creation of the general equilibrium theory

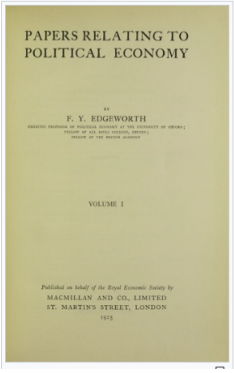
上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

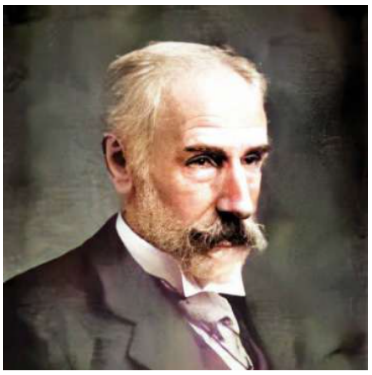
一般均衡同时考虑多个市场。若价格是 p，把总需求与总供给之差记为 $z(p)$，市场出清可抽象成求 $z(p)=0$。这只是说明数学问题的形式；本页没有给出 z 的具体模型或存在证明。

9 原课件第 9 页：Edgeworth

Introduction







Francis Ysidro Edgeworth (1881), **Mathematical Physics**

February 8, 1845 - February 13, 1926

Mathematical Physics: An Essay on the Application of Mathematics to the Moral Sciences (1881)

The Economic Journal (1891)

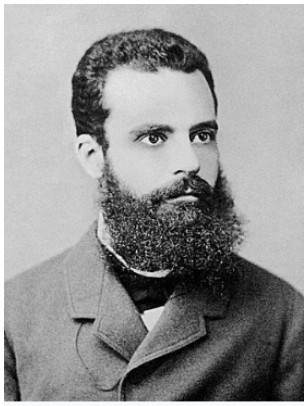
Edgeworth's Limit Theorem

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

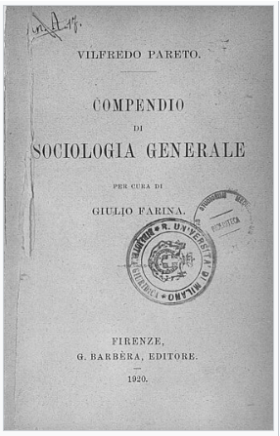
本页列出人物、著作和极限定理的名称。“极限”描述一族对象在某种变化下的趋近行为，后面从数列开始学习。不要凭名称猜测定理的假设与结论；考试引用结果时要写完整适用条件。

10 原课件第 10 页：Pareto 与效率

Introduction



Pareto in the 1870s





Vilfredo Federico Damaso Pareto (1896)

- July 15, 1848 - August 19, 1923
- French-Italian sociologist, economist and philosopher
- Pareto Efficiency / Pareto Optimality / Pareto Improvement

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

若一种可行改变使至少一人更好、无人更差，称为 Pareto 改进；不存在这种改进的可行配置称为 Pareto 最优。它不意味着人人同样富，也不自动意味着存在支持它的均衡价格，后者需要凸性和分离工具。

11 原课件第 11 页：von Neumann

Introduction



von Neumann in the 1940s





John von Neumann (1928, 1944)

- December 28, 1903 - February 8, 1957
- Mathematician, Economist, Computer Scientist, and so on
- Minimax Theorem (1928)
- Theory of Games and Economic Behavior (1944)

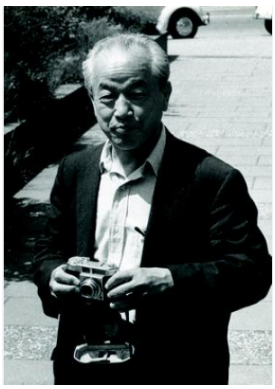
上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

极小极大思想研究两个决策者目标相反时的策略选择。最优化结果经常需要紧性保证最优点存在，凸性允许混合。这里先了解用途，等学完107、109再理解其数学基础。

12 原课件第 12 页：Kakutani

Introduction







Shizuo Kakutani (1941)

- August 28, 1911 - August 17, 2004
- Japanese and American mathematician
- Kakutani's Fixed-Point Theorem (1941)

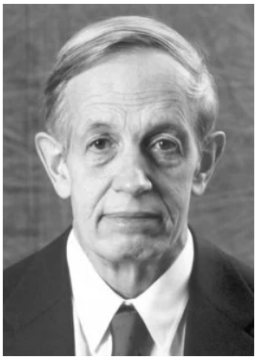
上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

对应把一个输入送到一组可能输出。不动点的形式是 $x \in \Phi(x)$，意思是当前选择本身属于允许的响应集。110将给出非空、紧、凸和半连续等条件，不能只背“不动点一定存在”。

13 原课件第 13 页：Nash

Introduction





John Forbes Nash Jr (1950)
- June 13, 1928 - May 23, 2015
- American mathematician
- Nash Equilibrium (1950)
- Kuhn - Tucker

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

均衡可以解释为每个人的选择都是对其他人选择的最佳响应。把所有最佳响应组合成对应，就能把均衡存在转为不动点问题。Kuhn–Tucker 条件属于优化的另一套工具；本页提到名称不等于要求先掌握它的证明。

14 原课件第 14 页：Arrow

Introduction



Kenneth Joseph Arrow (1950s)

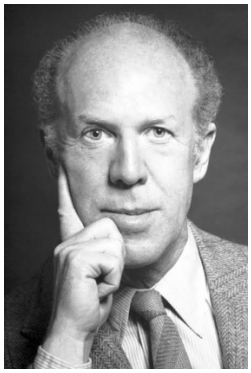
- August 23, 1921 - February 17, 2017
- American economist, mathematician and political throeist
- Social Choice Theory (Arrow's Impossibility Theorem)
- General Equilibrium Theory
- Fundamental Theorems of Welfare Economics
- Endogenous-Growth Theory
- Information Economics

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页涉及社会选择、一般均衡、福利、增长和信息等研究领域。不同领域的定理假设差别很大。学习方法是区分“某人的研究主题”和“本课程已建立的数学命题”，不从人物介绍推出未经证明的结论。

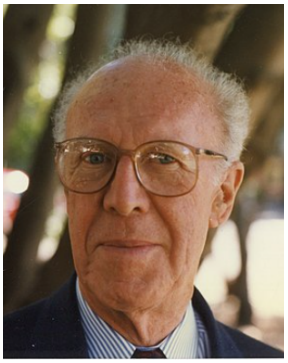
15 原课件第 15 页：Debreu

Introduction





Debreu in 1977



Gerard Debreu (1954)

- July 4, 1921 - December 31, 2004
- French-born economist and mathematician
- General Equilibrium Theory (1954)

Kenneth J. Arrow and Gerard Debreu (1954), *Existence of an Equilibrium for a Competitive Economy*, Econometrica 22, 265 - 290

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本课采用其公理化思路：先明确对象与性质，再从假设推出结论。例：要证明连续函数在集合上取最大值，必须补充集合非空且紧等条件；只写“函数连续”不够。

16 原课件第 16 页：人物年表汇总

Mathematical Economics

1. Adam Smith: June 5 (OS) / June 16 (NS) 1723 - July 17, 1790
2. Antoine Augustin Cournot: August 28, 1801 - March 31, 1877
3. Marie-Esprit-Leon Walras: December 16, 1834 - January 5, 1910
4. Francis Ysidro Edgeworth: February 8, 1845 - February 13, 1926
5. Vilfredo Federico Damaso Pareto: July 15, 1848 - August 19, 1923
6. John von Neumann: December 28, 1903 - February 8, 1957
7. Shizuo Kakutani: August 28, 1911 - August 17, 2004
8. John Forbes Nash Jr: June 13, 1928 - May 23, 2015
9. Kenneth Joseph Arrow: August 23, 1921 - February 17, 2017
10. Gerard Debreu: July 4, 1921 - December 31, 2004

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页重复汇总十位人物，前页的历史信息都保留在原图中。建议把学习笔记的时间用于记录“这个概念用来解决什么问题”，数学主线是：可行集合、最优选择、连续响应、均衡不动点。

17 原课件第 17 页：另一份经济学家名单

10 Economists Chosen by MIT Department of Economics

1. Adam Smith: June 5 (OS) / June 16 (NS) 1723 - July 17, 1790
2. David Richard: April 18, 1772 - September 11, 1823
3. Antoine Augustin Cournot: August 28, 1801 - March 31, 1877
4. John Stuart Mill: May 20, 1806 - May 8, 1873
5. Karl Heinrich Marx: May 5, 1818 - March 14, 1883
6. Marie-Esprit-Leon Walras: December 16, 1834 - January 5, 1910
7. Alfred Marshall: July 26, 1842 - July 13, 1924
8. John Maynard Keynes: June 5, 1883 - April 21, 1946
9. John von Neumann: December 28, 1903 - February 8, 1957
10. Joan Violet Robinson: October 31, 1903 - August 5, 1983

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页为历史名单。原文 David Richard 通常应为 David Ricardo（大卫·李嘉图）；这是人物名称说明，不影响后续数学。此页没有数学命题需要证明。

18 原课件第 18 页：二十世纪人物

2 Great Economists in Century 20



1. Paul Anthony Samuelson: May 15, 1915 - December 13, 2009
2. Kenneth Joseph Arrow: August 23, 1921 - February 17, 2017

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

本页突出 Samuelson 与 Arrow；“Great”属于课程介绍的评价性表述。我们的学习任务仍是理解严谨推导，而不是据此给研究者排序。没有新增数学假设、公式或证明。

19 原课件第 19 页：四种存在性方法

Introduction

There are FOUR distinct, but closely related, approaches to the existence problem of general equilibrium.

1. Fixed-Point Theorem [Brouwer (1911) and Kakutani (1941) and Analogous Arguments]
2. Efficient Algorithms — Approximate Economic Equilibrium [Scarf (1967)]
3. Fixed-Point Index + Degree Theory [Dierker (1972, 1974); Mityagin (1972); Smale (1974); Balasko (1975, 1978); Varian (1975); Nishimura (1978); Kalman and Lin (1978); and Kalman, Lin and Wiesmeth (1978)]
4. Differential Process — Generic Convergence [Smale (1976)]

上图为课件原页，保留原文、公式、图形及可能的笔误；下文为自学讲解。

不动点方法是本组讲义的主要目标；算法方法研究近似解；指数和度理论研究映射的整体结构；微分过程研究演化。存在、唯一、可计算、收敛是不同问题。例如 $f(x)=x$ 在区间上有无穷多个不动点：存在并不推出唯一。